DiffBHAR: training BHAR's acquisition policy with gradients

Results summary

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In one paragraph

BHAR tunes a two-parameter send rule per sensor with pattern search on a proxy reward (active-region fraction). We make BHAR's sensor simulator differentiable and train the policy, a small per-sensor neural network, directly on the classification loss of the held prediction. Everything else is BHAR's own code: data, splits, cached TinyHAR, finetuning and test. **On 3 of 4 datasets we match BHAR's accuracy with 14–39% fewer packets**; Opportunity (4 subjects) is worse. Giving BHAR's search 16× the budget changes neither its accuracy nor its packets. Under BHAR's harsher setting (half the harvested energy) all policies tie. The estimated on-sensor cost of the network is below the energy saved on all 4 datasets.

1 Method

Differentiable simulator. Forward pass identical to BHAR's `sparsify_data` (packet for packet). Backward: straight-through estimator for the send decision, a sigmoid surrogate for the energy-feasibility gate ($E_k \geq E_{\mathrm{Th}}$), and $-E_{\mathrm{Th}}$ for the energy a packet uses.

Policy. Every 200 ms each sensor computes, from harvested energy only (reading the accelerometer would cost more than the sensor harvests),

$$u_k = \left[\tilde{E}_k,\ \tilde{P}_k,\ p_k^{\mathrm{short}},\ p_k^{\mathrm{short}} - \tilde{P}_k,\ \Delta_k / f_s\right] :$$

stored energy, harvest rate over 1 s and over 200 ms (a free proxy for motion, since harvesting is kinetic), their difference, and time since the last packet. A per-sensor MLP gives the send logit, $\ell_k = w_2^\top \tanh(W_1 u_k + c_1) + c_2$ with 16 hidden units: 113 parameters, ∼100 multiply–adds per decision. BHAR's rule, send if $\Delta_k \geq \theta_e(1 - \tilde{E}_k)\tilde{P}_k + \theta_p$, uses a subset of these inputs.

Loss. Cross-entropy of the held prediction at every time step, i.e. the total (active + passive) error that BHAR's reward approximates. The classifier is frozen during policy training.

Exploding gradient (found and fixed). When the rule binds, each packet's time depends on the previous one through the reset of Δ; the backward pass multiplies by $\approx 0.05(\Delta + 9)$ per packet, about 10 at $\theta_p = 200$, so gradients reached 10^{10}–10^{20} with random sign. Stopping the gradient through the reset (forward unchanged) gives correct gradient directions ($t = 2$–8 on real data).

Training (per test subject and seed, BHAR's training/validation sequences): (1) the network imitates the fixed rule $\theta = (0, 100)$ (chosen once on PAMAP2, never re-tuned); (2) 16 epochs × 64 random 100 s segments, Adam; (3) keep the policy with the lowest validation loss (the start included); (4) BHAR's finetuning and test, unchanged.

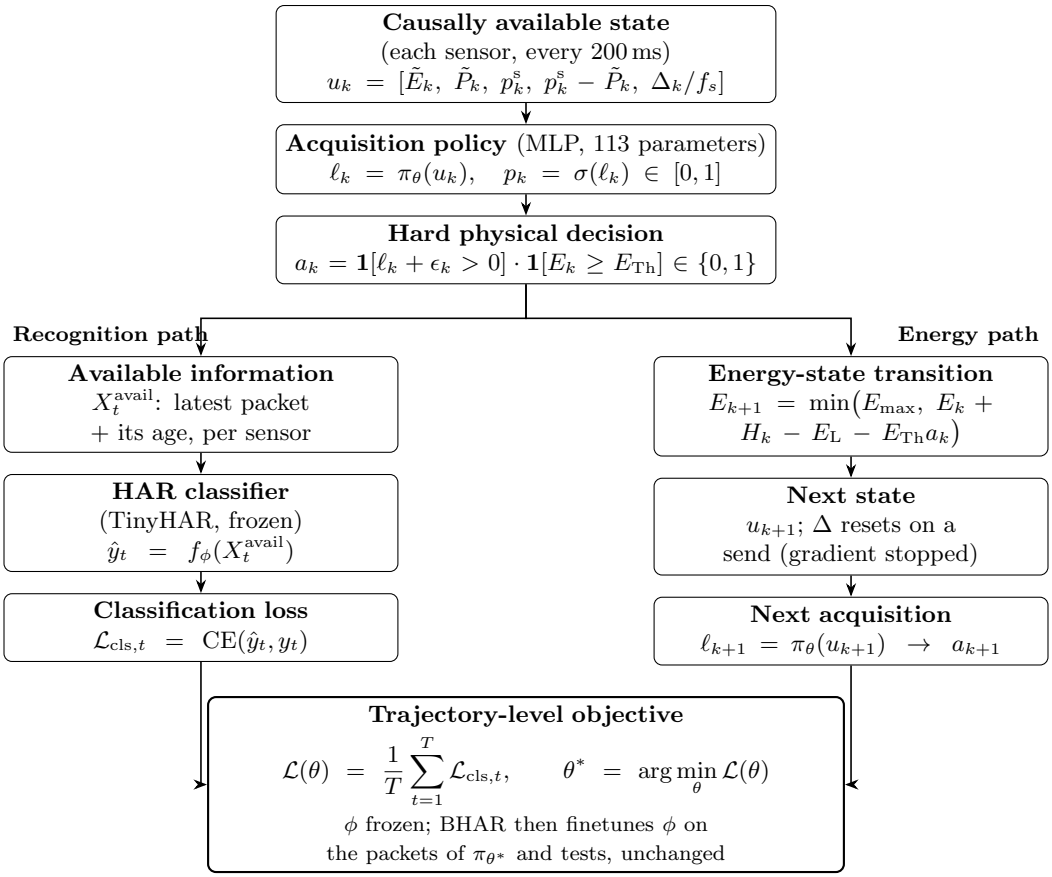


Figure 1: Training pipeline. Forward pass: identical to BHAR's sensor simulator (hard decisions, packet for packet). Backward pass: straight-through estimator for a_k (logistic noise ϵ_k), a sigmoid surrogate for the feasibility gate, $-E_{\mathrm{Th}}$ for the energy a packet uses. The state uses only the capacitor voltage (stored energy $\tilde{E}$, harvest rate over 1 s $\tilde{P}$ and 200 ms p^{s}) and the time since the last packet Δ.

2 Results

Leave-one-subject-out, 3 seeds, BHAR's protocol (same-subject validation split). BHAR is our rerun of BHAR's code (within seed noise of the authors' cached numbers; opportunistic matches exactly).

Table 1: Test accuracy / F1 / packets per second per sensor. Fixed (0,100) is our starting rule without training.

	PAMAP2	DSADS	RWHAR	Opportunity
Opportunistic	.510 / .454 / .261	.547 / .526 / .257	.614 / .636 / .332	.575 / .537 / .139
BHAR	.573±.004 / .522 / .196	.575±.007 / .564 / .178	.638±.013 / .653 / .255	.581±.023 / .544 / .123
Fixed (0,100), no training	.574±.005 / .532 / .137	.577±.016 / .566 / .147	.625±.012 / .638 / .145	.574±.007 / .560 / .108
Ours	**.578**±.005 / **.531** / **.144**	**.586**±.012 / **.574** / **.153**	.635±.016 / .647 / **.155**	.555±.020 / .541 / **.111**
Ours vs. BHAR (acc / packets)	+0.5 / −27%	+1.1 / −14%	−0.3 / −39%	−2.6 / −9%

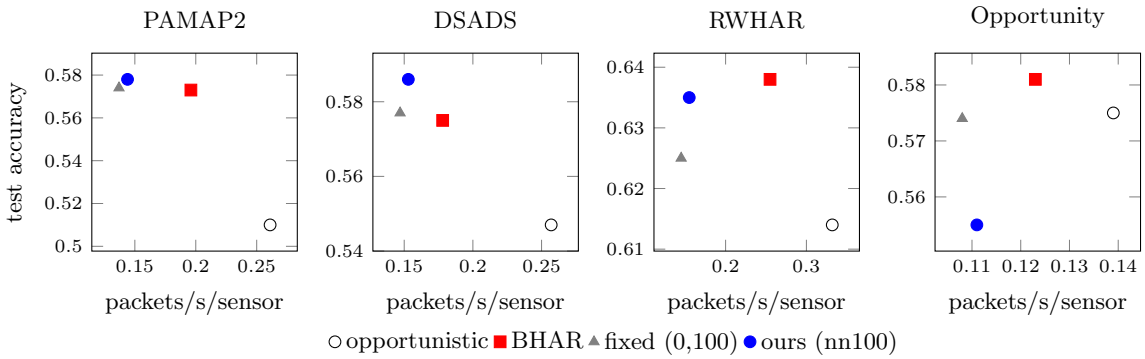


Figure 2: Accuracy vs. packets (up and left is better). Our policy sits left of BHAR at equal or higher accuracy on PAMAP2, DSADS and RWHAR.

Training itself helps. Over the untrained start: PAMAP2 +0.4, DSADS +0.9 (all 8 subjects improve), RWHAR +1.0 (8 of 11). The start rule carried over from PAMAP2 is below BHAR on RWHAR and Opportunity; training brings it to BHAR's level or above on 3 of 4 datasets without per-dataset tuning.

Not a compute advantage. BHAR's search sees 62 segments per sensor, ours 1024. With 512 search epochs (1024 segments) BHAR gets .574 / .525 / .200 packets on PAMAP2, unchanged from .573 / .522 / .196.

Scarcer energy (BHAR's sensitivity setting). Harvesting efficiency 0.5→0.25 on PAMAP2 (BHAR's Fig. 11d), everything else unchanged; BHAR re-searched, our recipe and start rule not re-tuned. Accuracy / F1 / packets: BHAR .446 / .413 / .094, fixed (0,100) .447 / .410 / .099, ours .449 / .413 / .101 (3 seeds). All three tie and are close to opportunistic (.110 packets): with half the energy almost every affordable packet has to be sent, so no policy has room to choose, and our packet saving disappears. The gain of a learned policy is limited to moderate energy budgets.

Deployment cost (estimate). The policy's inputs come from the capacitor-voltage readings BHAR already takes, so no extra sensing. We compiled one decision of the 16-unit MLP for the nRF52832 (Cortex-M4F, 64 MHz; `gcc -O3`) and counted cycles; at 3.7 mA and 3 V (datasheet, running from flash) a cycle costs ≈0.17 nJ. Float with a rational tanh: about 1,000 cycles, **0.17 µJ per decision**; 8-bit weights with a tanh lookup table: about 530 cycles, **0.09 µJ**; weights take 452 B (float) or 113 B (8-bit). BHAR's rule is 10 instructions (negligible). For comparison, Learned Sampling (BSN 2025) measures 0.2 µJ per policy evaluation on the same MCU.

The network only matters when a packet is affordable ($E_k \geq E_{\text{Th}}$); otherwise the sensor compares the voltage with the threshold, as BHAR does, and skips it. Replaying our policies on the test data, this is the case at 50% (PAMAP2), 51% (DSADS), 58% (RWHAR) and 31% (Opportunity) of the 200 ms steps. Against the packets saved ($E_{\text{Th}} = 50\,\mu\text{J}$), in µW per sensor:

	PAMAP2	DSADS	RWHAR	Opportunity
Packets saved vs. BHAR (/s/sensor)	.052	.024	.101	.012
Energy saved	2.62	1.21	5.04	0.58
Network runs (fraction of steps)	50%	51%	58%	31%
Policy cost, float, every step	0.90	0.90	0.90	0.90
Policy cost, float, only if $E_k \geq E_{\text{Th}}$	0.45	0.45	0.52	0.27
Policy cost, 8-bit, only if $E_k \geq E_{\text{Th}}$	0.22	0.23	0.26	0.14
Net, float, every step	+1.72	+0.31	+4.14	−0.32
Net, 8-bit, only if $E_k \geq E_{\text{Th}}$	**+2.40**	**+0.98**	**+4.78**	**+0.44**

Cycle counts are estimates from the compiled code, not measurements, and the policy cost is not in the simulator (the gated 8-bit version would lower the harvest rate by about $0.2\,\mu$W, against $E_{\mathrm{L}} = 6.6\,\mu$W). Whether 8-bit weights change any decision is still to be checked. Total sensor power ($6.6\,\mu$W idle + packets + policy) is 10% (PAMAP2), 2% (DSADS), 21% (RWHAR) lower than BHAR's and 2% higher on Opportunity with the ungated float policy; the main gain is fewer radio transmissions at equal accuracy.

Limitation: Opportunity. Only 3 training subjects per fold; training makes 3 of 4 test subjects worse (worse timing, same coverage), i.e. it overfits the training subjects. Two regularisers (more orderings of the training data; a KL penalty towards the start) did not fix it. BHAR is only 0.6 points above opportunistic there, within its own seed spread.

Side findings. (i) Simple fixed rules dominate BHAR on PAMAP2: $\theta = (0, 100)$ has the same accuracy with 30% fewer packets. (ii) BHAR's same-subject validation split favours subject-sensitive policies (it cannot see rules that collapse to opportunistic on a new subject), which limits what any policy optimiser can gain under this protocol.

3 Questions

1. Is this worth a paper as it stands (equal accuracy, 14–39% fewer packets, no gain under scarce energy)?
2. Venue: a 4-page letter now (IEEE Embedded Systems Letters), or DAC (Nov 18) / CODES (late March) first with these results?
3. How should Opportunity be handled, and is its spread (BHAR $\pm.023$, cache .598 vs. our rerun .581) known?